BOOK REFERENCE

FORMULA REFERENCE

Every core formula in one place, straight from the game's balance config.

UPDATED 2026-10-09

PROGRESSION

Level threshold (cumulative XP):

$$\mathrm{xpForLevel}(n) = \left\lfloor 5.32 \cdot (n-1)^{2.6986} \right\rfloor \qquad \mathrm{xpForLevel}(1) = 0$$

You are the highest level whose threshold you have crossed. Each level up pays **3 gold × the level reached**. The daily reward grants **10 XP**, and so does each daily mission. The XP bar displays progress within the current level and resets on level up.

Golden Hours run every day from **17:00 to 18:00 UTC**: work shifts and war strikes pay **double XP** inside the window. The daily reward and mission payouts stay flat.

ENERGY

Maximum energy by level (regen is **+1 per minute**):

$$E_{\max}(\mathrm{level}) = \min\left(300,\ 100 + 5\,(\mathrm{level}-1)\right)$$

The region's Health index, a Conservative government and a convoy Field Hospital speed the clock, their shares added together (see the theory). During Golden Hours energy regenerates twice as fast; only the minutes inside the window count double.

Food adds energy on top of regen: **10 / 20 / 30 / 40 / 50** per unit for T1–T5, at most **10 units per UTC day** (20 with Premium), and never above your maximum. Energy drinks cost **0.10 gold** per energy point; one use fills your whole deficit, and a **10-minute** cooldown separates uses.

WORK

Units produced per block of energy, before the region's effects:

$$u_0 = 2 \cdot \left(1 + 0.02\,S_{\text{eco}} + 0.015\,S_{\text{edu}} + 0.005\,(\text{level} - 1)\right) \cdot m_{\text{company}} \cdot \left(1 + \tfrac{b_{\text{region}}}{100}\right)$$

$$u = \max\left(1,\ u_0 \cdot m_{\text{industry}} \cdot m_{\text{crowd}} \cdot m_{\text{pioneer}}\right)$$

b_{region} is the region's resource bonus. It counts only for a Resource Plant that produces the region's own resource.

$m_{\text{industry}} = 1 + 0.25 \cdot \frac{\text{industry}}{10} \cdot \frac{\text{electricity}}{10} \cdot \text{crown} + \text{ideology} + \text{convoy}$ is the region's Industry index (scaled by its electricity), plus a flat +0.15 under Socialism and up to +0.10 from a Mobile Workshop in your convoy. The index and the ideology are those of the company's region and the country that owns it.

m_{crowd} and m_{pioneer} come from the region's shared work capacity (see Infrastructure & Region Capacity):

- m_{crowd} is **1** while the region's energy pool is at least half full. Below that, output slides down as the pool empties, to **×0.5** at the very bottom.
- m_{pioneer} is **1.10** while **15 or fewer** players have been active in the region in the last 24 hours. It fades to 1 at 30 active players.

Here **crown** is 1.30, 1.20 or 1.10 in the three cities ranked highest by that day's City Score, and 1 everywhere else. A city's City Score is the average of its five indexes, rounded down; the cities are ranked every day at 00:00 UTC. Ideology, convoy and Premium are never multiplied by the crown.

Units in one shift (each block costs **10** energy):

$$\text{units} = \left\lfloor \frac{\text{energy} \cdot u \cdot m_{\text{capital}}}{10} \right\rfloor$$

m_{capital} is **0.9** for 30 days after the company's country lost its capital in war, and 1 otherwise.

Nothing is rounded before this last step, so every bonus counts, however small. A +10% resource bonus lifts a rate of 6.6 units per block to 7.26, and a 100-energy shift then makes 72 units instead of 66.

The shift charges only the energy its whole units needed, rounded up; the leftover stays in your bar. A factory short of raw material, a plant short of an input, or a share worker's full storage shortens the shift the same way: you pay only for what was made.

Company multiplier m_{company}:

$$m_{\text{RAW}} = 1 + 0.01\,(\text{level} - 1), \qquad m_{\text{MFG}} = \text{tierMult} \cdot \left(1 + 0.02\,(\text{level} - 1)\right)$$

m_{RAW} applies to Resource Plants. Factories, Light Industry and Shipyards use m_{MFG}. **tierMult** is the tier yield, **1.0 / 0.8 / 0.65 / 0.5 / 0.4** for T1–T5; Light Industry always works at the T1 yield.

Raw material per manufactured unit is **10 / 20 / 35 / 60 / 100** for T1–T5; tickets take **25 / 50 / 90 / 150 / 250** oil instead. A Light Industry component takes **5 + 5** raws. A ship takes construction materials, electronics and fuel worth the raw of four weapons of its tier. A plant with several inputs stops at the scarcest one.

Work XP: **4 per 10 energy spent** (×2 in Golden Hours), fractions dropped.

TRAINING

$$\text{cost}(n) = n \text{ gold}, \qquad \text{time}(n) = \text{round}\left(0.2894 \cdot n^{2.5} \cdot \frac{1}{1+0.01\,S_{\text{edu}}}\right) \text{ seconds}$$

The time is never under **one second** on this curve. It is then cut by:

- the region's Education index: up to **–25%**, times the crown;
- Liberalism: a flat **–15%**;
- a convoy Field Academy: up to **–10%**;
- Premium: **–10%**.

The cuts add up, and a training never drops below a tenth of its time. Only one stat trains at a time.

TAX

$$\text{tax} = \lfloor \text{amount} \cdot \text{rate} \rfloor \quad (\text{money taxes: minimum } 0.01 \text{ on any positive taxed amount})$$

Rates are clamped to **0–50%**. The production tax hits **Resource Plant output only**, is paid in kind into the state's resource store, and has no minimum.

Who pays what:

- **Income tax** — on wages, and on the seller's proceeds on the local market.
- **Import tax** — the buyer's country rate, added on top of the global-market price. An active trade agreement between the two countries sets it to zero, in both directions.
- **Export tax** — the seller's country rate, taken from the seller's proceeds.
- **Frontier levy** — a side with no country pays a flat **10%** in place of its import or export tax. It is burned; no treasury receives it.

Rates change through a **Set a tax rate** law: parliament votes on it, or the president or the Economy Minister enacts it directly.

WAR

Weapons burned per 10 energy (level-scaled):

$$w_{10} = 1 + \left\lfloor \frac{\text{level}}{8} \right\rfloor$$

Damage of one strike (all current energy is spent, minimum 10, in 10-energy blocks rounded up — $\lceil \text{energy}/10 \rceil \cdot w_{10}$ weapons):

$$\text{damage} = \text{weapons} \cdot (5 + S_{\text{str}}) \cdot m_{\text{tier}} \cdot \left(1 + b_{\text{military}} + b_{\text{nationalism}} + b_{\text{FC}}\right) \cdot m_{\text{capital}}$$

$b_{\text{military}} = 0.20 \cdot \frac{\text{military}}{10} \cdot \frac{\text{electricity}}{10} \cdot \text{crown}$ is the Military index bonus of the region you strike from; $b_{\text{nationalism}} = 0.10$ when the state that **owns that region** is Nationalist (your citizenship does not matter); b_{FC} is your convoy's Forward Command (up to +0.08). These are added into one multiplier. Only m_{capital} multiplies separately: 0.9 for 30 days after your country loses its capital, else 1.

Tier multiplier m_{tier}: **1.0 / 1.5 / 2.0 / 2.5 / 3.0** for T1–T5. Fight XP: **20 per 10 energy** (×2 in Golden Hours).

Every hit is then rolled: a uniform swing of ±10%, and a 5% chance of a critical at ×1.5. Only the rolled number is rounded down:

$$\text{dealt} = \max\left(1,\ \lfloor \text{damage} \cdot s \cdot c \rfloor\right), \qquad s \sim U(0.9,\ 1.1), \qquad c = 1.5 \text{ with } p = 0.05, \text{ else } 1$$

If you hold fewer weapons than your energy calls for, the strike spends only the energy your weapons cover, in whole blocks; the rest stays in your bar.

Naval landing. In the landing phase of a naval war, ships replace weapons: w_{10} ships per **40** energy, in whole 40-energy blocks only (minimum 40). Each ship counts as **four weapons** of its tier in the damage formula, so a landing deals the same damage per energy as a weapon strike.

Wars last **24 h**; ties go to the defender. A naval war has two 24-hour phases. The landing comes first. If the attackers finish it strictly ahead, the assault opens, fought with weapons, and the attackers start it with **half** the landing's margin. A tied or lost landing ends the war, and nothing is razed.

When a war ends, the losing side's region loses between **10% and 50%** of each kind of building: the closer the war, the more it loses.

STORAGE

Each item has its own storage limit. The base is **750** for raws, **150** for components, **150 / 90 / 52 / 30 / 15** for food, weapons and tickets (T1–T5) and **30 / 18 / 10 / 6 / 3** for ships. It grows **+15%** per level above 1, **+3%** per Strength point and **+2%** per Economy point, and the total is rounded down.

FLIGHT

$$\text{minutes} = \text{round}\big(10 + \text{km} \cdot 0.0235 \cdot f_{\text{tier}}\big)$$

Tier factor f_{tier}: **1.0 / 0.75 / 0.55 / 0.38 / 0.25** for T1–T5.

IDEOLOGY

A state ideology adds a flat share on one axis in every region of the state, independent of the indexes and of electricity:

$$m = 1 \pm \big(\text{region term} + \text{ideology}\big), \qquad \text{ideology} = \text{its share on its axis, else } 0$$

The shares: Nationalism 0.10, Socialism 0.15, Liberalism 0.15, Conservatism 0.25.

Nationalism → damage, Socialism → industry, Liberalism → training time (subtracted), Conservatism → health regen.

The bonus follows the country that owns the region: the region you work in for production, the region you stand in for regen and training, and the front you strike from for damage.

← PREVIOUS
WAR & THE FRONT

NEXT →
THE THEORY OF THE PARLAMENTUM ECONOMY